Discover Computing

Research

Opinion mining using ensemble model for restaurant feedback analysis

Rohan S. Kamath[1] · M. R. Kaushik[1] · M. Ramakrishna[1]

Received: 9 May 2024 / Accepted: 29 April 2025
Published online: 18 June 2025

Abstract
The research work introduces a new method for sentiment analysis in the context of restaurant evaluations, emphasizing the optimization of the trade-off between processing time and accuracy. A majority voting classifier is used in our research to evaluate and merge multiple machine learning models, such as Decision Tree, Support Vector Machine, Random Forest, Logistic Regression, K-Nearest Neighbors, Multinomial Naive Bayes, and an Ensemble Method. Thorough preparation of the data, feature extraction, and analysis are all part of the study. Models are then trained and evaluated using performance metrics like accuracy, precision, recall, and F1 score. The main contribution of this work is the integration of models with optimized hyperparameters into an ensemble framework, which greatly improves model performance. Our results show that the ensemble method-the combination of Support Vector Machine, Logistic Regression, and Multinomial Naive Bayes-performs better than both single and multiple models in terms of accuracy and efficiency. Thus, it is determined that the ensemble approach is the best way to perform sentiment analysis on restaurant evaluations, giving restaurant owners valuable business knowledge and important insights for raising customer happiness. By demonstrating the enhanced performance and usefulness of an improved ensemble model in actual food and beverage industry settings, this study contributes to the field of sentiment analysis.

Keywords Classification models · Ensemble · Machine learning · Processing time · Restaurant reviews · Opinion mining

1 Introduction

Opinion mining (OM), also known as sentiment analysis (SA), is a computational approach used to analyze individuals' opinions and emotions towards diverse subjects. By soliciting feedback from customers, it is possible to conduct a comprehensive analysis of the data obtained, thus facilitating the advancement and expansion of a corporate enterprise. Feedback provides valuable and perceptive recommendations for enhancing growth and progress. It enables the comprehension of the areas in which individuals, groups, or entities excel, as well as the areas that necessitate enhancements or modifications [1]. The Internet is more popular as a source of information, leading to a rise in the utilization of sentiment analysis to assess public sentiment. Sentiment analysis is employed within the gastronomy sector to evaluate client comments and evaluations that are published on diverse websites and portals [2] [3].

Rohan S. Kamath and M. R. Kaushik have contributed equally.

✉ M. Ramakrishna, ramakrishna.m@manipal.edu; Rohan S. Kamath, rohankamath11@gmail.com; rohan10.mitmpl2022@learner.manipal.edu; M. R. Kaushik, kaushikmendon65@gmail.com; kaushik3.mitmpl2022@learner.manipal.edu | [1]Department of Data Science and Computer Applications, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India.

Client feedback plays a crucial role in guaranteeing client happiness within company enterprises. The comprehension of client requirements and the effective resolution of their concerns have the potential to foster heightened levels of customer loyalty. Customers can express their thoughts, emotions, and evaluations using a range of techniques, such as written reviews, comments, social media posts, polls, surveys, and other means of communication. The authenticity and use of client feedback render it increasingly significant. The work exemplifies authentic and unaltered sentiments that remain unaffected by extraneous influences. The feedback, regardless of its positivity, negativity, or neutrality, accurately reflects the genuine opinions of customers who have used a product or service in real-life situations. Moreover, the inclusion of feedback from organizations or enterprises in the evaluation of their products or services creates a possible source of bias and inaccuracy inside the feedback loop, hence giving rise to various concerns. When organizations or enterprises engage in the process of offering product reviews and feedback, they must practice prudence to uphold openness and authenticity. The presence of biased or erroneous reviews has the potential to erode trust, impede significant change initiatives, and ultimately harm a company's relationships with its stakeholders and customers. Hence, organizations or enterprises must prioritize the utilization of objective, transparent, and customer-centric feedback channels to uphold their credibility and foster authentic growth.

Machine learning is a highly suitable approach for sentiment analysis in the context of enterprises, researchers, and organizations that aim to extract insights from client feedback and other sources. Lexicon-based methods rely on sentiment lexicons, which are lists of words or phrases that have been assigned negative and positive ratings, and also calculate the sentiment of the text based on these ratings. Common dictionaries such as AFINN and VADER are often used for this purpose [4] [5]. Machine-learning-based methods use algorithms and models to perform sentiment analysis. Supervised sentiment analysis uses labeled data sets to train models such as Naive Bayes, Support Vector Machines, Random Forest [6] or even deep learning models such as Convolutional Neural Networks (CNN) [7] and Networks recurrent neurons (RNN) [8]. Unsupervised learning methods do not require labeled data and often use techniques such as clustering. Latent Dirichlet Allocation (LDA) [9] and K-means clustering are examples of unsupervised methods. Hybrid methods offer a unique combination of lexicon-based and machine-learning techniques, combining the strengths of both to improve accuracy and performance. These methods can initially use sentiment lexicons and then refine the results using machine learning. Rule-based systems, on the other hand, rely on predefined rules and language models to distinguish sentiment. These rules can be based on grammatical structures, language patterns, or domain-specific knowledge. Aspect-based sentiment analysis delves deeper into text by examining sentiment at the aspect or attribute level. It is especially useful for distinguishing how different components of a product or service are perceived, as evidenced by restaurant reviews that rate food, service, and atmosphere separately.

In recent years, deep learning methods have gained prominence in sentiment analysis, especially recurrent neural networks (RNNs) and transformation models such as BERT and GPT [10] [11]. These models capture complex contextual information, making them highly effective for natural language understanding tasks. The choice of sentiment analysis method depends on factors such as available data, domain-specific requirements, and the level of precision required in sentiment assessment.

This work focuses on customer feedback in restaurants using sentiment analysis and machine learning models. The goal is to develop supervised learning models for classifying feedback into positive and negative feelings. The methodology includes data collection, pre-processing, and the use of various machine learning classification models, which are well-suited for this analysis. Also, we have used hyperparameter tuning, which meticulously fine-tunes the hyperparameters of the machine learning models used, aiming to optimize their performance and accuracy. This fine-tuning process ensures that the models are operating at their highest potential and capable of capturing nuanced aspects of customer sentiment.

This study also considers the ensemble of machine learning models. By combining multiple models, it leverages the diverse strengths and perspectives of each, ultimately enhancing the accuracy of the model by using hyperparameters. The ensemble model seeks to mitigate the weaknesses of individual models, providing a more comprehensive understanding of customer feedback. Overall, this study aims to conduct a comparative exploration of machine learning models, also ensemble all the classification models to give the best accurate result by considering their best processing time so that it provides restaurant owners and managers with tools that can swiftly and accurately decode customer sentiments, enabling prompt responses and improvements to enhance customer satisfaction.The uniqueness of this article is the comprehensive comparison and evaluation of various machine learning models for sentiment analysis of restaurant customer reviews, and the use of an overall technique to achieve higher processing accuracy and efficiency. This is due to the strategic combination of individual models.

1.1 Contribution

The novelty of this article is that it provides a comprehensive comparison of sentiment analysis techniques used by different researchers to analyze customer reviews in the hospitality industry. It focuses on machine learning techniques to identify attitudes expressed in textual data, such as positivity and negativity.

This study further delves into methodologies such as data preprocessing, feature extraction, classification model construction, model evaluation, hyperparameter tuning, combination selection, and majority voting classification. It also provides insight into the value of customer feedback in improving customer satisfaction and business prospects for restaurateurs. Additionally, this article presents a detailed analysis of sentiment analysis in restaurant reviews using machine learning models and ensemble techniques to achieve high accuracy and efficient processing time. Comparing different classification models and ensemble techniques provides valuable insight into the best approach for sentiment analysis in restaurant reviews.

Overall, this article contributes to a better understanding of opinion mining related to restaurant reviews and provides a structured approach and comprehensive analysis in this field.

2 Literature Review

This review compares sentiment analysis techniques used by different researchers to analyze customer feedback in the hospitality industry, focusing on machine learning methods to identify attitudes expressed in textual data, such as positivity, negativity, and neutrality.

In their respective papers, Medhat et al. [2] and Wankhade et al. [1] have intensively dealt with the complex area of sentiment analysis. Medhat et al. provided a detailed explanation of sentiment analysis and defined it as the computational processing of beliefs, feelings, and subjectivity in text data. They also conducted a review of various sentiment analyses and finally concluded that Support Vector Machine (SVM) and Naive Bayes are the most commonly used techniques among all other models.However, they did not explore the possibility of enhancing performance by merging various machine-learning models. We filled this gap in our work by using ensemble approaches, which integrate the advantages of several different classification models. By using this method, sentiment analysis model robustness is increased along with predicted accuracy [2].

Wankhade et al. [1] further discussed sentiment analysis by not only providing a comprehensive explanation but also expanding their exploration to include additional supervised learning methods such as Logistic Regression, Maximum Entropy and K Nearest Neighbour. This approach highlights their commitment to thoroughly exploring the landscape of sentiment analysis methodologies. However, they did not concentrate on adjusting the hyperparameters of these models to maximize their performance. Hyper-parameter tuning for every classification model is part of our study. In order to maximize the models' performance and guarantee the best possible sentiment categorization accuracy, this step is essential.

Patil et al. [12] used a combination of supervised and unsupervised machine learning techniques for sentiment analysis. Specifically, they used the Naïve Bayes approach, a proven supervised classification model, as well as K-means Clustering, an important unsupervised classification technique. Their analysis was conducted using a data set consisting of restaurant reviews collected from various social media platforms. Furthermore, the authors also used data mining to extract information from the data set. The advantage of their approach is that the application of K-means clustering to classify restaurant reviews. Using this technique, the authors tried to give customers a better understanding of a restaurant's features and offerings based on their preferences. This clustering approach gives customers access to categorized information that can help them make informed decisions about food choices. However, they did not take advantage of ensemble learning techniques to efficiently merge these models. By combining the results of several models to increase the overall accuracy and dependability of the sentiment analysis, we were able to outperform their strategy by utilizing ensemble learning techniques [12].

Bhavitha et al. [13] conducted a study on sentiment analysis, based on reviews and customer attitudes towards new products. Their research included a comparison of different modeling approaches, specifically evaluating machine learning and lexicon-based methods in different data sets. Among the different models examined, the Machine Learning approach, which primarily uses a Support Vector Machine (SVM), provided the highest accuracy of 86.4%. This result shows the effectiveness of SVM in sentiment analysis tasks, particularly in the specific context of product reviews and customer attitudes. Additionally, their research came to a conclusion regarding model selection based on data set size.

They found that Naive Bayes tends to perform well when working with smaller data sets, while SVM performs perfectly when working with larger data sets. These findings provide practical guidance for sentiment analysis practitioners and provide a nuanced understanding of which approach to use based on the scope of available data.They did not investigate how to incorporate these models into a unified ensemble framework. We expanded on their research by including SVM, Naive Bayes, and more models into a group framework. By combining the advantages of both models, we can produce a sentiment analysis which is more accurate [13].

Rahate et al. [14] shed light on the substantial influence of user-generated content on social media platforms. They highlighted how this content serves as a crucial means for individuals to express their emotions and opinions and highlighted the growing need for automated sentiment analysis, facilitated by natural language processing techniques. The authors highlighted the relevance of sentiment analysis in the context of deciphering user sentiment and decision-making processes revolving around product ratings and online interactions.The use of automated sentiment analysis based on natural language processing plays a crucial role in this endeavor. By leveraging these advanced techniques, researchers and businesses can gain a deeper understanding of user sentiment and the factors that influence their decision-making processes, thereby improving their ability to make informed decisions and interact effectively with their customers. However, they did not offer a specific strategy to improve the precision and effectiveness of these analyses. In order to address this, we created a thorough procedure that uses ensemble models, hyper-parameter tuning, and data preprocessing. Our sentiment analysis is accurate and efficient because of this thorough approach [14].

Krishna et al. [15] conducted comparative analysis on various Supervised Classification Machine Learning Models, which include Naïve Bayes, Support Vector Machine (SVM), Decision Tree, K-Nearest Neighbours (KNN), and Random Forest. Their study aimed to assess the performance of these models in the context of a specific data set and task. They systematically built and evaluated these machine learning models using three different data splits i.e., 70-30,75-25,80-20, One of the noteworthy findings from their study was that the Support Vector Machine (SVM) model outperformed the other models in terms of accuracy. Specifically, in the 80-20 data split, where 80% of the data was allocated for training purposes and 20% for testing purposes, the SVM model achieved an accuracy of 94.56%. They did not investigate the advantages of using ensemble methods to aggregate the advantages of each model. We combine models such as Naive Bayes, SVM, Decision Tree, K-Nearest Neighbors, and Random Forest with ensemble approaches in our study. The sentiment analysis's overall accuracy and robustness are enhanced by this combination [15].

Laksono et al. [16] analyzed by extracting data from TripAdvisor, a popular travel review platform. To gather this data, they used the WebHarvy tool, which allowed them to extract information from the platform. They also harnessed the power of Weka 3.8.2, a data mining tool, to analyze and extract useful information from this data set. They constructed two distinct models for their analysis namely, Naïve Bayes and Text Blob. These models were designed to perform sentiment analysis on the extracted data. Overall, the results of their study indicated that Naïve Bayes outperformed Text Blob in terms of accuracy. The Naïve Bayes model achieved the best accuracy of 72.06%, while Text Blob exhibited slightly lower accuracy, with a score of 69.12%. This finding suggests that Naïve Bayes was the more effective model for sentiment analysis within the context of their research. However because the models were so basic, the study was unable to handle complex emotions and inconsistent evaluations. We used numerous models to improve predicted performance in our work using ensemble learning techniques, which helps overcome these restrictions. More specifically, we combined Random Forest, SVM, and Naïve Bayes models to increase overall accuracy [16].

Shamantha et al. [17] authors have delved into a detailed comparative analysis of various supervised machine learning models, namely Naïve Bayes, Support Vector Machine (SVM), and Random Forest. Their data set consists of 156,061 data points, which were mined from Twitter using hashtags, reflecting the vast array of user-generated content on the platform. Through their comparative analysis, they arrived at an important result that the SVM model was the best performing among the various models examined. SVM achieved an impressive accuracy rate of 78.3% during its experiments. This convincing result led to the conclusion that SVM is not only better but also suitable for processing large data sets, underlining its ability in sentiment analysis tasks involving significant amounts of data. They did not integrate these models into an ensemble framework to enhance performance. Instead of solely focusing on individual models, we combined the best-performing models into an ensemble framework. This integration allowed us to leverage the strengths of each model, leading to improved predictive accuracy in sentiment analysis [17].

Zahoor et al. [18] conducted a comparative analysis of several supervised machine learning classification models, including Naïve Bayes, Support Vector Machine (SVM), Random Forest, and Logistic Regression. The main objective of their study was to evaluate and compare the performance of these different machine-learning models in a specific data set. They conducted a systematic study of these machine learning models on three different data splits, i.e., 70-30, 75-25, and 80-20 respectively. Furthermore, they classified the data set into four different categories, namely Food Taste, Service, Value for Money, and Atmosphere, reflecting the complexity of the sentiment analysis task that they approached. Their results show that the Random Forest model achieved an impressive 95% accuracy in both the 75-25 and 70-30 splits of the data. This result shows the effectiveness of Random Forest in handling sentiment analysis within this specific data set and different splitting criteria. Their study's narrow feature set and concentration on a particular dataset might not fully capture the complexities of restaurant reviews. Our research tackled this limitation by employing feature extraction techniques to extract more comprehensive and meaningful features from the reviews. We also utilized advanced preprocessing methods to handle various text formats and improve model performance [18].

Sharif et al. [19] began a comparative analysis of several supervised machine learning models, particularly Naïve Bayes, Decision Trees, and Random Forest. Their study included a data set of 1,000 data points in the Bengali language. To evaluate and compare the performance of these classification models, they used K-fold cross-validation, a technique used for evaluating machine learning models. Their result shows that the Naïve Bayes model achieved the highest accuracy of 80.48%. This finding highlights the effectiveness of the Naïve Bayes approach in performing sentiment analysis in the context of their Bengali language data set. This study was limited by the 1,000 data point dataset, which might not have allowed for a reliable assessment of the model's performance. We addressed this limitation by expanding our dataset within the same source to include more data points, ensuring a more comprehensive evaluation of our ensemble model [19].

Huda et al. [20] conducted a comparative analysis of various supervised machine-learning classification models. Their analysis includes models such as Naive Bayes, Support Vector Machine (SVM), Logistic Regression, and Decision Trees. To ensure the reliability of their analysis, they manually collected the data set. Additionally, they used two different feature extraction methods, namely Count Vectorizer and TF-IDF. According to their results, the SVM model supported by TF-IDF feature extraction achieved the highest accuracy rate, an impressive 95% [20].

Gunawan et al. [21] conducted a survey using data from TripAdvisor, a well-known travel review platform. The objective of their study was to use sentiment analysis techniques to understand the data available on this platform better. To achieve this, they evaluated two different models, which include Naive Bayes and Support Vector Machine (SVM). These models have been carefully designed and trained to analyze the sentiments in the data set. The results show that the Support Vector Machine (SVM) model outperformed Naïve Bayes. Specifically, the SVM model achieved an accuracy of 79%. This study did not capture the whole range of opinions expressed in restaurant reviews. Our study expanded the data sources to include multiple review platforms, thereby capturing a broader range of sentiments and improving the generalizability of our findings. By leveraging ensemble methods, we were able to achieve higher accuracy and robustness across diverse datasets [21].

A. Alrehili and K. Albalawi [22] have collected a data set consisting of 34,661 records, each containing 21 features, which includes customer reviews, prices, and other relevant data. This research used three classification models: Naive Bayes, Support Vector Machine (SVM), and Random Forest. These models are trained on the data set to make predictions. Furthermore, the study used two ensemble models Bagging and Boosting. To further increase the predictive power, the authors used a majority voting algorithm. This ensemble model combined the results of individual classification models and ensemble models and made predictions based on the majority of decisions of the other models. The research found that Random Forest achieved the highest accuracy among the models with a score of 89.87%. This result shows that Random Forest outperforms other models in classifying electronic products, making it a promising approach for tasks such as product categorization and recommendation systems. This study was constrained by the computing expense and complexity of ensemble approaches. To solve this problem, we optimized our ensemble model with fast algorithms and parallel processing, which allowed us to reduce computing costs without sacrificing accuracy. Our method made sure that the advantages of group learning were achieved without sacrificing scalability or performance [22].

Mamun et al. [23] have developed a Bangali Sentiment Analysis data set (BSaD), by collecting from various Bangla language sources including online news portals, Facebook posts/comments, YouTube comments, and blogs. The goal of this data set is to facilitate sentiment analysis specifically in the Bengali language to meet the needs of a large language community. The study highlights the application of five different classification models to the BSaD data set. These models include widely used logistic regression, decision trees, random forest, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). Each of these models serves as a tool to predict the sentiments expressed in Bengali textual data. A notable aspect of the research is the integration of an ensemble model, specifically AdaBoost, into the analysis. AdaBoost, a popular ensemble technique, merges the predictions of multiple weak learners to create a more robust and accurate model. By using this ensemble method, the research aims to further refine the predictive performance of each classification model, thereby improving the overall sentiment analysis capabilities for Bengali texts. The research result shows that the combination of Logistic Regression (LR), Random Forest (RF), and Support Vector Machine (SVM) provides the highest accuracy among the models used. The reported accuracy of 82% highlights the success of this model ensemble in sentiment analysis for Bengali texts. This result suggests that the combined strength of LR, RF, and SVM, perhaps due to their different approaches and strengths, results in a powerful sentiment analysis system tailored to the Bengali language. While their integration of AdaBoost enhanced the model's robustness, they faced issues with overfitting due to the diverse sources of Bengali text and the varying writing styles. To overcome these problems, we combined models like Naive Bayes, Support Vector Machine, and Logistic Regression into a more thorough ensemble approach. Using the advantages of each model, this strategy reduced overfitting and increased accuracy all around [23].

Oussous et al. [24], started with data collection which consists of reviews and comments gathered from various Arabic-language social media platforms. The research consists of two distinct experiments. The first experiment involved a comparison of different classification models such as Naive Bayes, Support Vector Machine (SVM), and Maximum Entropy to ascertain their respective performance in the context of Arabic-language sentiment analysis. The SVM model achieved an accuracy rate of 82.5%. The second experiment focuses on using ensemble methods, such as majority voting and stack generalization. In this phase of the research, the majority voting algorithm demonstrated its effectiveness by achieving the highest accuracy of 83.45%. Moreover, it delivered these results in a short time frame, requiring only 31.78 s. Stack generalization, another ensemble approach was evaluated using the Naive Bayes algorithm. It also delivered results with an accuracy rate of 83.15%, with a longer processing time of 379.43 s. These findings show that ensemble methods outperformed individual classification algorithms in terms of accuracy. Their research, however, was hampered by the noise that exists naturally in social media data, such as language, acronyms, and multilingual content which resulted in less than ideal outcomes. We addressed the challenge of noisy data by implementing advanced preprocessing techniques to clean and normalize the text data. However, we did not use social media datasets, we applied these techniques to our dataset to ensure high-quality inputs [24].

S. Al-Saqqa et al. [25] conducted a sentiment analysis in the Arabic language data sets. The research was based on the use of three different data sets, each offering a perspective on Arab sentiment. The first data set, the Opinion Corpus of Arabic (OCA), was used as the primary text data source. The second data set comes from the UCI repository. The third data set, the Large-Scale Arab Sentiment Analysis (LABR) data set, contributed to the overall scope of the research. To analyze these data sets, four different machine learning classification models were applied such as Naive Bayes, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Decision Trees (DT). Along with these individual models, an ensemble technique like majority voting, was used to combine different combinations of models and evaluate their effectiveness. The research results showed that the performance of these models and ensembles was different across the three data sets. For the first data set, the Combination of Naïve Bayes SVM and DT models got the highest accuracy of 86.8%, For the second data set, SVM got an accuracy of 84.4%. For the third data set, the Combination of Naïve Bayes, SVM, and DT models provided an accuracy of 85.94%. Their approach was limited by the complexity of Arabic morphology and the scarcity of annotated datasets. These challenges resulted in moderate accuracy levels and difficulties in generalizing the model across different Arabic dialects. We addressed similar challenges by focusing on effective preprocessing techniques and selecting robust machine-learning models. We ensured that our dataset was well-annotated and applied preprocessing steps to handle morphological complexities [25].

Carrasco, P., and Dias, S. [26] implemented an Aspect-Based Sentiment Analysis (ABSA), an automated approach that uses sophisticated natural language processing (NLP) techniques to analyze consumer attitudes in restaurant reviews. With ChatGPT obtaining the highest F1 Score, it demonstrates the BART model's greater performance over the DeBERTa model. The study's limited broader application is due to its dependence on a 1000-review sample and its concentration on the Algarve region. The main problem with the comparison research is that it relies too heavily on a small dataset of 1000 reviews, which limits the applicability of the results to bigger datasets and other locations outside of the Algarve. Our work, on the other hand, overcomes this problem by using a larger dataset of 1424 reviews, providing a more equitable representation of both positive and negative feelings. Furthermore, in order to improve forecast accuracy, we have combined different categorization models into an ensemble approach [26].

Gupta, V., and Rattan, P. [27] started with pre-processing, feature extraction, unsupervised classification techniques, and aspect-based sentiment analysis to forecast restaurant survival. Classification approaches include FCM, K-Means, and HBPSO-Optimized FCM, while feature extraction uses TF-IDF vectors, Ngram, Bag of Words, and Word Embedding. The results of the evaluation demonstrate the efficacy of this strategy; HBPSO-Optimized FCM had the maximum accuracy of 89.50%. These results highlight how crucial user-generated material is to restaurant sector decision-making. The paper focuses on aspect-based sentiment analysis using unsupervised methods like Fuzzy C-Means and K-Means clustering to predict restaurant survival. This approach has limitations, such as the potential for reduced accuracy in distinguishing between nuanced customer sentiments and reliance on unsupervised methods that may not always capture the full complexity of the data. In contrast, our paper addresses these issues by employing a combination of supervised and ensemble learning techniques, leveraging hyperparameter tuning and model ensembling to improve accuracy and performance [27].

Table 1, shows the Comparative analysis of contributions and limitations of existing sentiment analysis research papers, along with our proposed paper.

Hence, In this work, the plan is to use ensemble models to address the challenges.

3 Proposed methodology

3.1 Problem statement

In today's world, the restaurant business relies on customer reviews or feedback, which provides information regarding restaurant services. The goal of this research is to enhance restaurant profitability and customer experience through the application of machine learning, specifically in classification and ensemble models.

3.2 Methodology

The methodology outlines the different steps involved in performing a restaurant review Opinion Mining, from data set collection, Building and calculating the performance of individual classification models, finding the best accurate model, and performing hyper-parameter tuning with different variations of parameters in each model.

Additionally, the methodology extends to the incorporation of ensemble methods, which entail combining the strengths of different classification models to enhance predictive accuracy. Moreover, the methodology takes into account the processing time of both individual models and ensembled models.

Figure 1, shows the flow chart representing steps to perform opinion mining starting from Data Collection till Evaluating the results.

3.2.1 Data Collection and Data Visualization

In this study, restaurant review sentiment analysis relies on a data set obtained from Kaggle, a platform for data science and machine learning resources [28]. This data set comprises two essential features such as customer reviews and sentiments, which are categorized into positive and negative classes. These labels are crucial in classifying the sentiments expressed within the customer reviews.

In Fig. 2, we can see that the data set consists of 1424 rows, among these reviews, 710 are categorized as positive, reflecting satisfaction and contentment, while 714 reviews fall into the negative category, indicating customer dissatisfaction.

3.2.2 Data preprocessing and analysis

Data pre-processing steps are important to clean and prepare review text data before performing sentiment analysis. Each step improves the quality and accuracy of the model. In this study, we have used some of the data preprocessing techniques, which are discussed here.

Initially, the drop method of the data object was used to identify and eliminate null and empty columns from the data set. After this stage, the data set was more comprehensive and organized, successfully eliminating any columns with empty or missing values. Subsequently, the data set was transformed into binary values for the sentiment analysis process. Using `LabelEncoder()`, this conversion gave the text data binary labels, usually representing "Positive" as 1 and "Negative" as 0, to help with sentiment analysis. The `data. isnull().sum()` function was used to detect missing values and any gaps identified in the data set. The `data.duplicated().sum()` function was used in identifying duplicates, which were subsequently removed. Furthermore, it involved removing special characters, punctuation, and symbols from reviews, thus cleaning up the text and preparing it for later analysis.

To ensure uniformity and consistency, the entire review text has been converted to lowercase using `review. lower()` function. This approach eliminated potential case-sensitivity issues during analysis, ensuring that all text data was treated equally. Tokenization was another crucial step in word processing. Each review was broken down

Table 1 Comparative analysis between existing paper and our paper

	Author's Contribution	Limitations/Gaps	Our Contribution
[2]	- Defined sentiment analysis as processing beliefs, feelings, subjectivity in text. - Reviewed various sentiment analysis techniques. - Concluded SVM and Naive Bayes are widely used.	- Did not explore possibility of combining multiple models.	- Used ensemble approaches to increase accuracy and robustness.
[1]	- Discussed sentiment analysis comprehensively. - Explored additional supervised methods like Logistic Regression, Maximum Entropy, KNN.	- Did not focus on hyperparameter tuning for models to maximize the performance.	- Included hyperparameter tuning to maximize model performance and guarantee the best possible sentiment categorization accuracy.
[12]	- Used both supervised and unsupervised machine learning techniques for sentiment analysis. - Used Naive Bayes and K-means clustering for sentiment analysis on restaurant reviews. - Highlighted restaurant features through K-means clustering.	- Did not utilize ensemble methods.	- Applied ensemble learning to increase overall accuracy and reliability.
[13]	- Compared machine learning and lexicon-based methods for product review sentiment analysis. - Found SVM effective for large datasets with accuracy of 86.4% and Naive Bayes for small datasets.	- Did not incorporate models into an ensemble framework.	- Combined SVM, Naive Bayes, and others in an ensemble for higher accuracy. - By combining the advantages of both models, we can produce a sentiment analysis that is more accurate
[14]	- Emphasized importance of user-generated content for sentiment analysis. - Applied NLP for decision-making support in product ratings.	- Did not suggest strategies to improve analysis accuracy.	- Developed an ensemble-based approach with hyperparameter tuning for higher accuracy and Efficiency.
[15]	- Conducted a comparative analysis of supervised classification models like Naïve Bayes, SVM, Decision Tree, KNN, Random Forest. - Evaluated performance on a specific dataset. - Achieved highest accuracy with SVM at 94.56% using an 80-20 data split.	- Did not explore ensemble methods to leverage strengths of individual models.	- Utilized ensemble techniques by combining Naïve Bayes, SVM, Decision Tree, KNN, and Random Forest. - Enhanced overall accuracy and robustness in sentiment analysis.
[16]	- Extracted data from TripAdvisor using WebHarvy. - Built Naïve Bayes and Text Blob models using Weka 3.8.2. - Naïve Bayes achieved higher accuracy at 72.06% compared to Text Blob's 69.12%.	- Basic models failed to capture complex emotions and inconsistencies.	- Employed ensemble learning techniques with Random Forest, SVM, and Naïve Bayes. - Improved predictive performance.
[17]	- Analyzed Naïve Bayes, SVM, and Random Forest on 156,061 Twitter posts. - SVM was the top performer with an accuracy of 78.3%.	- Did not implement an ensemble framework to enhance performance.	- Integrated best-performing models into an ensemble framework. - Leveraged strengths of individual models to improve predictive accuracy.
[18]	- Analyzed Naïve Bayes, SVM, Random Forest, and Logistic Regression on a dataset categorized by various aspects (Food Taste, Service, etc.). - Random Forest achieved the highest accuracy of 95% on 70-30 and 75-25 splits.	- Narrow feature selection might not capture the full complexity of restaurant reviews.	- Employed advanced feature extraction techniques. - Improved model performance with advanced preprocessing methods.

Table 1 (continued)

	Author's Contribution	Limitations/Gaps	Our Contribution
[19]	- Analyzed Naïve Bayes, Decision Trees, and Random Forest on a Bengali dataset of 1,000 points. - Naïve Bayes achieved an accuracy of 80.48%.	- Limited by a small dataset, affecting reliability of performance assessment.	- Expanded dataset significantly for a more robust evaluation.
[20]	- Analyzed supervised classification models like Naive Bayes, SVM, Logistic Regression, Decision Trees. - Used a manually collected dataset. - Applied Count Vectorizer and TF-IDF for feature extraction. - Found SVM with TF-IDF achieved highest accuracy at 95%.	- Did not explore ensemble methods or advanced preprocessing techniques.	- Built on their findings by incorporating ensemble techniques and advanced preprocessing methods. - Aimed to improve sentiment analysis accuracy and robustness.
[21]	- Conducted a survey using TripAdvisor data for sentiment analysis. - Evaluated Naive Bayes and SVM models. - Found SVM outperformed Naive Bayes with an accuracy of 79%.	- Did not capture the full range of opinions in restaurant reviews.	- Expanded data sources to include multiple review platforms. - Captured a broader range of sentiments and improved generalizability. - Leveraged ensemble methods for higher accuracy across diverse datasets.
[22]	- Collected a dataset of 34,661 records with 21 features (customer reviews, prices). - Applied Naive Bayes, SVM, Random Forest, and ensemble techniques (Bagging, Boosting). - Random Forest achieved the highest accuracy at 89.87%.	- Constrained by computing expense and complexity of ensemble approaches.	- Optimized ensemble model using fast algorithms and parallel processing. - Reduced computing costs without sacrificing accuracy or performance.
[23]	- Developed the Bengali Sentiment Analysis dataset (BSaD) from various sources (online news, social media). - Applied classification models: Logistic Regression, Decision Trees, Random Forest, KNN, SVM, and AdaBoost. - Achieved highest accuracy at 82% with a model ensemble.	- Encountered overfitting issues due to diverse text sources and styles.	- Combined Naive Bayes, SVM, and Logistic Regression in a comprehensive ensemble approach. - Reduced overfitting and improved overall accuracy.
[24]	- Collected reviews from various Arabic-language social media platforms. - Conducted two experiments: • First experiment compared Naive Bayes, SVM, and Maximum Entropy models. • Found SVM achieved an accuracy of 82.5%. - Second experiment focused on ensemble methods: • Majority voting algorithm yielded highest accuracy of 83.45% in 31.78 seconds. • Stack generalization using Naive Bayes achieved 83.15% accuracy but took 379.43 seconds. - Results demonstrated ensemble methods outperformed individual algorithms.	- Hampered by noise in social media data (language, acronyms, multilingual content). - Resulted in less than ideal outcomes.	- Addressed noisy data challenge with advanced preprocessing techniques. - Applied techniques to our dataset for high-quality inputs.

Table 1 (continued)

	Author's Contribution	Limitations/Gaps	Our Contribution
[25]	- Analyzed Arabic datasets from OCA, UCI repository, and LABR. - Used Naive Bayes, SVM, KNN, Decision Trees, and majority voting. - Highest accuracy was 86.8% for Naive Bayes, SVM, and DT on OCA dataset. - SVM achieved 84.4% on UCI dataset; combination reached 85.94% on LABR dataset.	- Limited by Arabic morphology complexity and scarce annotated datasets. - Resulted in moderate accuracy and difficulties in model generalization.	- Focused on effective preprocessing techniques and robust ML models. - Ensured well-annotated dataset and applied preprocessing for morphology.
[26]	- Implemented an Aspect-Based Sentiment Analysis (ABSA) approach using advanced NLP techniques. - ChatGPT achieved the highest F1 Score, outperforming DeBERTa. - Study limited by reliance on 1,000 reviews from the Algarve region.	- Comparison research limited by small dataset of 1,000 reviews. - Reduces applicability of results to larger datasets and different locations.	- Used a larger dataset of 1,424 reviews for more equitable representation. - Combined different categorization models into an ensemble approach to improve forecast accuracy.
[27]	- Employed preprocessing, feature extraction, unsupervised classification, and aspect-based sentiment analysis to predict restaurant survival. - Utilized classification methods like Fuzzy C-Means (FCM), K-Means, HBPSO-Optimized FCM. - Applied feature extraction techniques such as TF-IDF vectors, N-grams, Bag of Words, Word Embedding - Achieved highest accuracy of 89.50% with HBPSO-Optimized FCM, highlighting the importance of user-generated content in restaurant decision-making. - Emphasized the use of unsupervised methods for aspect-based sentiment analysis to forecast restaurant survival.	- Reduced accuracy in distinguishing customer sentiments. - Reliance on unsupervised methods may not capture the full complexity of the data.	- Employing a combination of supervised and ensemble learning techniques. - Leverages hyperparameter tuning and model ensembling to improve accuracy and performance.

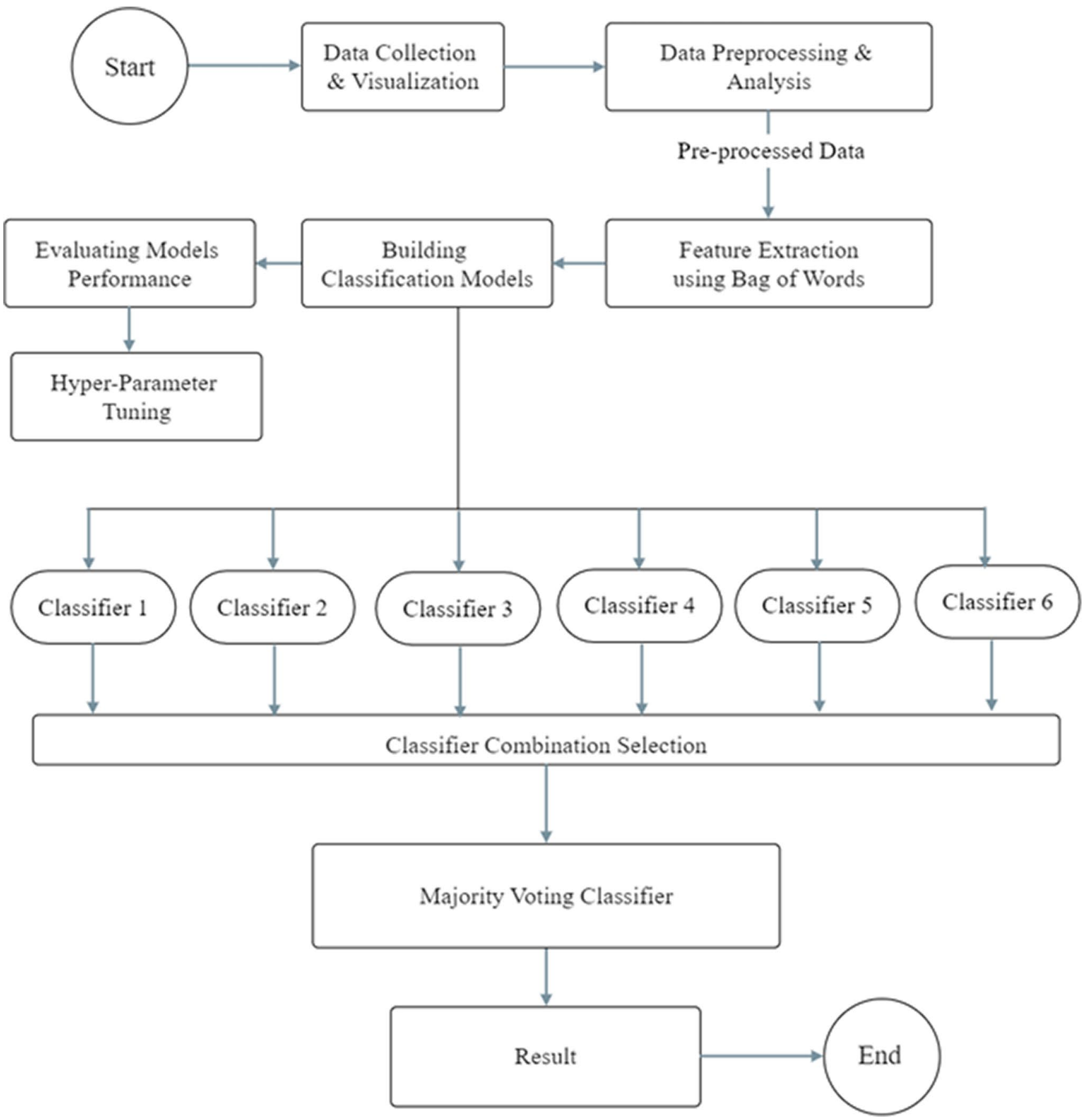


Fig. 1 Proposed methodology flow chart

into individual words using `review.split()`, essentially creating tokens for each word. These tokens served as fundamental units for further analysis, making it easier to perform.

Finally, words were extracted, a process in which words were reduced to their most basic forms or roots using `PorterStemmer()` function. For example, words like "running" have been converted to their root form "run." This transformation made it easier to collect related word variants, thereby improving the analysis of textual data.

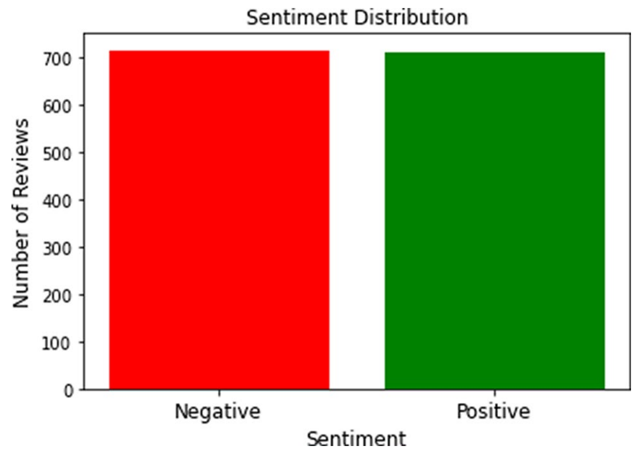


Fig. 2 Distribution of sentiments in the dataset

3.2.3 Feature extraction

In this research, we have utilized the Bag of Words (BoW) method using Count Vectorization techniques as a fundamental part of the model for sentiment analysis. The key to implementing this method is by giving the pre-processed data, The Bag of Words model is a text representation technique that treats data as an unordered series of words, ignoring grammar and word order. Creates a vocabulary of unique words from the entire data set and counts the frequency of each word in a document, creating a numeric vector representing the document. This vector is called a "bag" because it stores a collection of words regardless of their order. These vectors are memory-efficient as they store mostly zeros.

It efficiently handles large datasets and does not require understanding grammatical structures, making it suitable for multilingual texts. Despite the high dimensionality and loss of word context, BoW's sparse representation can handle it effectively. Combining BoW with other methods, such as TF-IDF weighting, can adjust word importance to improve performance. Although BoW has limitations, its advantages make it a widely used and effective method in many applications.

Count Vectorization is a technique to implement the Bag of Words method. It Converts a collection of text documents into an array of token counts. As shown in Fig. 3, each row in the matrix corresponds to a document, and each column corresponds to a unique vocabulary word. The values in the matrix represent the number of times each word appears in each document.

Let n be the number of documents in the corpus and m be the size of the vocabulary; then the count vectorization matrix X can be defined as:

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,m} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,m} \end{bmatrix}$$

The key features of count vectorization used in this methodology are *n_grams* i.e. range as (1,2), which means that it can have both single words (*unigrams*) as well as pair of words (*bigrams*) and max-features, which will help in managing dimensionality of data in the process.

3.2.4 Building classification models

Model building is a crucial step in sentiment analysis, where various classifiers are trained on the training data to learn input features and their labels. The resulting patterns can then be used to predict the meaning of the unseen data

Fig. 3 Feature extraction vector

	Word 1	Word 2	……..	Word N
Document 1	1	0	……..	1
Document 2	0	1	……..	0
……	……	…….	……..	…….
Document N	1	0		0

in the test set. In this research, we have used classification models such as Decision Tree, Multinomial Naïve Bayes, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Random Forest, and Logistic Regression.

The data set was strategically divided into two different sets: the training set and the testing set. This division plays a crucial role in machine learning because it allows models to be trained on a specific part of the data. This research followed a common practice of splitting the data in a 70:30 ratio, with 70% of the data for training and the remaining 30% for testing. The research consists of several machine learning classifiers.

1. Decision Tree First, the Decision Trees classifier (CART) was used which, after rigorous training on the provided data, generated a model that resembled a tree structure [29]. The general form of the Decision Tree structure

$$T(x) = \begin{cases} \text{Leaf node value}, & \text{if at leaf node} \\ T_{\text{left}}(x), & \text{if condition is true} \\ T_{\text{right}}(x), & \text{if condition is false} \end{cases} \tag{1}$$

From Equation (1): $T_{\text{left}}(x)$ and $T_{\text{right}}(x)$ are the subtrees or next decision nodes based on the outcome of the decision condition. This model works as a decision-making tool and makes predictions based on attribute values.

2. Multinomial Naïve Bayes

The next classifier is Multinomial Naïve Bayes where Training is done using data from the training set to understand the probability of occurrence of words in the context of different sentiment classes. This probabilistic approach is a fundamental aspect of Naïve Bayes classifiers and plays an important role in predicting sentiment based on the presence of specific words. This algorithm can be applied in the text-based scenario by converting it to an integer countable nominal form. [30] The calculation of the probability is described in Equation (2):

$$P(c|d) \propto P(c) \prod P(t_k|c) \tag{2}$$

$P(t_k|c)$ is the conditional probability that the word appears in a document with class c. In the equation, $P(t_k|c)$, the likelihood probability of class c is t_k and P(c) is the prior probability. The purpose of class determination is to compare posterior probability results obtained and then the class with the largest posterior probability is chosen as the predicted result. The prior probability equation can be seen in the Equation (3):

$$P(c) = \frac{N_c}{N} \tag{3}$$

N_c is the sum of category c, while N is the sum of all categories. The likelihood probability Equation can be seen in the Equation (4):

$$P(t_k|c) = \frac{T_{kc}}{\sum_{t' \in V} T_{ct''}} \tag{4}$$

T_{kc} is the number of occurrences of the word t in the document having class c, and $T_{kc} \sum_{t' \in V} T_{ct''}$ is the total number of occurrences of all words in class c [30].

3. Support Vector Machine Support Vector Machine is another classification technique used. SVM excels in finding optimal hyperplanes that can divide data into different classes. By training on the provided data set, SVM finds the most appropriate hyperplane to differentiate between sentiments. [31]For a given training data set considering (x_i, y_i), where x_i is the input feature vector, and y_i is the class label (− or + for binary classification). The decision function for classifying a new data point $\mathbf{x}$ is given by:

$$f(\mathbf{x}) = \text{sign}(\mathbf{w} \cdot \mathbf{x} + b) \tag{5}$$

From Equation (5):

- $\mathbf{w}$ is the weight vector.
- $\mathbf{x}$ is the input feature vector.
- b is the bias term.
- sign is the sign function that returns −1 if the argument is negative, 0 if it's zero, and +1 if it's positive.

The hyperplane is determined by minimizing the following Equation (6):

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N} \max(0, 1 - y_i(\mathbf{w}\cdot\mathbf{x}_i + b)) \tag{6}$$

From Equation (6),

- $\|\mathbf{w}\|^2$ is the squared norm of the weight vector.
- C is a regularization parameter that controls the trade-off between maximizing the margin and minimizing the classification error.
- The sum term represents the hinge loss, which penalizes points that fall on the wrong side of the margin [31].

4. Random Forest This study also uses the Random Forest classification Model, which builds a large number of decision trees and merges their predictions to increase accuracy. The diversity and combination of multiple decision trees make it a better choice for complex classification tasks. [32] Considering Random Forest algorithm as RF. Given an input vector X, the prediction is made by $RF(X)$ is the aggregation of predictions from individual decision trees:

$$RF(X) = \frac{1}{N}\sum_{i=1}^{N} T_i(X) \tag{7}$$

From Equation (7):

- N is the number of decision trees in the forest.
- $T_i(X)$ is the prediction of the i-th decision tree.
- For classification tasks, the final prediction is often the class that receives the majority of votes among the individual trees.
- For regression tasks, the final prediction is often the average (mean) of the predictions from all trees.

Each tree T_i in the Random Forest is typically constructed using a subset of the training data and a subset of features, promoting diversity among the trees [32].

5. Logistic regression Logistic regression was also applied, this classifier builds a model that estimates the probability that belongs to a specific class. It uses the logistic function which is also called as sigmoid function, which forms the basis for probability estimation and provides information about the probability of a given sentiment. [33] Let's Consider logistic regression model as $f(X)$, where X is the input vector.The logistic function is defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{8}$$

From Equation (8):

- z is the linear combination of input features and their associated weights and bias term given by,

$$z = w_0 + w_1x_1 + w_2x_2 + \ldots + w_nx_n$$

The logistic regression model can be denoted as:

$$f(X) = \sigma(z) = \frac{1}{1 + e^{-(w_0 + w_1x_1 + w_2x_2 + \ldots + w_nx_n)}} \tag{9}$$

From Equation (9):

- $\mathbf{w}$ is the vector of weights
- $\mathbf{X}$ is the input vector
- $\sigma(\cdot)$ is the logistic function.

The logistic regression model is trained by finding the weights $\mathbf{w}$ that maximize the likelihood of the observed data [33].

6. K-Nearest Neighbors Finally, the K-Nearest Neighbors (KNN) classifier was used with its method to store all data points and class labels in the training set. When a new data point appears, KNN identifies the K nearest neighbors in the training set and assigns the input data point to the class that appears most frequently among these neighbors. [34] Considering KNN classifier as $KNN(X, k)$, where X is the input vector and k is the number of neighbors. The mathematical equation to predict the class C for a new observation X using Equation (10):

$$KNN(X, k) = \text{argmax}_C \left(\sum_{i=1}^{k} \delta(y_i, C) \right) \tag{10}$$

From Equation (10):

- y_i is the class label of the i-th nearest neighbor to X.
- $\delta(y_i, C)$ is the Kronecker delta function, which is one if $y_i = C$ and 0 otherwise.
- argmax_C finds the class C that maximizes the sum [34].

3.2.5 Evaluating models performance

Model Evaluation is a process of applying metrics such as accuracy, precision, recall, and F1-score to the test data to know each model's performance. Accuracy measures the proportion of correctly classified instances out of the total number of instances. It provides the overall accuracy of a model. Precision measures the ratio of truly positive predictions to the total number of instances that would be positive. Recall measures the ratio of true positive predictions to the total number of actual positive cases. It quantifies a model's ability to capture all positive examples. It focuses on the accuracy of positive predictions. The F1 score is the harmonic mean of precision and recall. It balances precision and recall, providing a single metric that reflects both.

Further, we have also calculated the processing time of each model. In this study, we have evaluated model performance using two techniques that are:

- *Confusion Matrix Plotting* Generate a confusion matrix to visualize the classifier's performance on the test set. True Positive (TP), where the model predicts Positive and the actual result is Positive. True Negative (TN), where the model predicts the Negative actual result is Negative. False Positive (FP), where the model predicts Positive and the actual result is Negative. False Negative (FN), where the model predicts Negative and actual is Positive. Overall, TP and TN should be greater than FP and FN; this results in that the model is best for classification.
- *Performance Metrics Calculation* To measure the performance of the classifier, the following evaluation metrics are used:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{12}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{13}$$

$$\text{F1 Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

3.2.6 Hyperparameter tuning

Hyperparameter tuning is a step to improve the performance of a machine learning model. It involves conducting experiments with combinations of various parameters to identify the ones that yield the most favorable outcomes.

Hyperparameter tuning is necessary as it directly affects model performance and accuracy. Proper tuning of hyperparameters such as learning rate, number of epochs, and batch size can improve the model's ability to learn from the data and prevent poor performance due to underfitting or overfitting. This balances outliers, reducing errors and improving the model's ability to generalize to unseen material. Moreover, carefully tuned hyperparameters can speed up convergence, save computational resources, and shorten development cycles. Effective tuning also mitigates overfitting, where the model learns noise from the training data, and underfitting, where the model is too simplistic to capture data patterns.

Firstly, the Decision Tree classifier underwent parameter tuning by changing `max_depth` value from 1 to 10. Several iterations were performed and the accuracy was calculated for each `max_depth` value. This method identifies the `max_depth` value that provided the best accuracy, to optimize the decision tree classifier.

The Naïve Bayes classifier was fine-tuned by adjusting the alpha parameter changing the values from 0.1 to 1.0. The study trained the classifier for each alpha value and measured its accuracy. This process identifies the best alpha value that corresponds to the highest accuracy, ensuring optimal performance of the model.

Similarly, the performance of the Random Forest classifier was optimized by changing the `n_estimators` parameter values from 50 to 200 in increments of 50. This approach shows the `n_estimators` value associated with the highest accuracy, thus achieving an optimal performance of the Model.

The Support Vector Machine (SVM) classifier was fine-tuned by changing kernel parameters, and values such as linear, rbf, poly, and sigmoid. SVM classifiers were trained for each kernel value and their respective accuracy was evaluated. This analysis allowed us to identify the best-performing kernel, providing information on the best parameter for SVM.

Logistic Regression was fine-tuned by changing the C parameter values to 0.001, 0.01, 0.1, 1, 10, and 100. This method trained the classifier for each C value calculated the accuracy, and determined the highest accuracy C value.

Lastly, the K-Nearest Neighbors (KNN) classifier was fine-tuned by varying the `n_neighbors` parameter within a range from 1 to 21. The classifier was trained for each `n_neighbors` value and the accuracy was calculated. This study identified the `n_neighbors` value associated with the best accuracy, ensuring optimal performance for the KNN classifier.

3.2.7 Combination selection

To increase sentiment classification performance and produce more accurate results, in this study, we have combined different combinations of classifiers with their best parameters by fine-tuning them. Each combination consists of different classifiers and a different number of combinations, as shown in Table 2. As detailed in Algorithm 1.

Every model illustrates a different method for establishing decision limits. When forming decisions, decision trees and random forests make use of tree structures, whereas support vector machines (SVMs) uses hyperplanes. Decisions are made using the K-Nearest Neighbors (KNN) method, logistic regressions describe probabilities linearly, and Naive Bayes is predicated on probabilistic assumptions. By combining these different models, This work achieve higher classification accuracy compared to relying on a single model approach.

Table 2 Combination Selection

Combination	Classifiers
C1	DT, NB,SVM, RF, LR, KNN
C2	NB, SVM, LR
C3	NB, SVM
C4	NB, LR

Algorithm 1 Combination Selection

```
procedure GenerateCombination
    Step 1: Initialization
    Step 2: Data Preparation, Model Training, and Evaluation
    Load_Train_Data()
    Split_Data_into_Train_Test()
    Initialize_Models()
    for each Model in Models do
        Build_Individual_Model(Model, Training_data)
        Train_Model(Model, Training_data)
        Fine_Tune_Model(Model, Training_data)
        Evaluate_Model(Model, Test_data)
    end for
    Step 3: Ensemble Building, Training, Evaluation, and Conclusion
    Combined_Models = Combine_Models(C1, C2, C3, C4)
    Output(Combined_Models)
    for each Ensemble_Model in [C1, C2, C3, C4] do
        Train_Ensemble_Model(Ensemble_Model, Training_data)
        Evaluate_Ensemble_Model(Ensemble_Model, Test_data)
    end for
    Compare_Individual_vs_Ensemble_Performance()
    Output_Final_Results()
end procedure
```

3.2.8 Majority voting classifier

A voting classifier is a machine-learning model that combines the predictions of different types of classifiers to make a final prediction. It is a form of ensemble learning, where combined predictions are generally more robust and accurate than predictions from individual classifiers. [35] Considering majority voting classifier as MV. Given N individual classifiers $C_1, C_2, \ldots, C_N$, the majority voting function for a binary classification is denoted by:

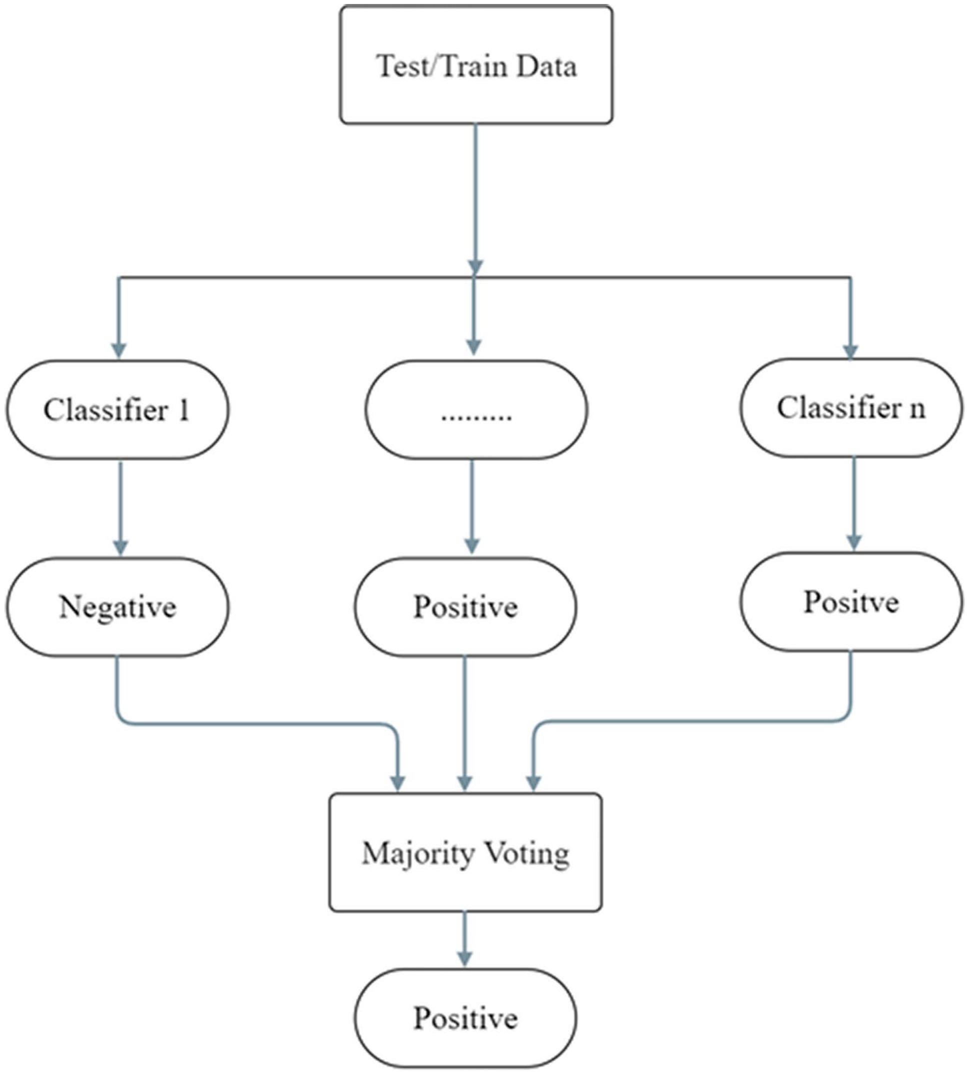
$$MV(X) = \text{sign}\left(\sum_{i=1}^{N} \text{vote}_{C_i}(X)\right) \tag{15}$$

From Equation (15):

- X is the input vector.
- $\text{vote}_{C_i}(X)$ is the vote (prediction) of the i-th classifier for the input X.
- $\text{sign}(\cdot)$ is the sign function, which outputs +1 if the argument is positive, 0 if it's zero, and -1 if it's negative [35].

In this study, we have used the hard voting technique (as shown in Fig. 4) in a majority voting classifier where each classifier in the ensemble predicts the class labels. The final prediction is determined by majority voting in which class that receives the most votes is the predicted class for the ensemble. This method works well when the individual classifiers have good performance.

Fig. 4 Hard Voting Classifier



4 Experiment and discussions

4.1 Implementation details

For this research on restaurant review Opinion Mining, the data set includes two features such as customer reviews and their corresponding sentiment labels, which are classified as either positive or negative. The data set consists of a total of 1,424 individual data. Among these data, 710 reviews have been labeled as positive and 714 reviews have been labeled as negative.

In this study, a software platform used is Anaconda Navigator, which serves as the central environment for managing various data science tools, packages, and libraries. Within the Anaconda Navigator environment, the IDE chosen for this study is Spyder. Spyder is a powerful and popular IDE designed specifically for machine learning tasks.

In this research, we have employed a variety of Python packages to facilitate the analysis and processing of textual data, as well as to build and evaluate machine learning models for Opinion Mining. For data manipulation and analysis, we used the Pandas library, which makes it easy to manage data in tabular form. NumPy is another important package for numerical operations and array manipulation. This research uses the Scikit-Learn library, which provides a wide range of machine learning tools, and also to create a sentiment classification model and evaluate its performance using various metrics including accuracy, precision, recall, and F1 score. To visualize our results and evaluate model performance, we use Matplotlib and Seaborn to create plots and visualizations, including the representation of the confusion matrix.

4.2 Results

The purpose of this experiment is to compare the accuracy of individual classification models and also find the most accurate model by using Hyperparameter tuning and assembling different combinations of classification models.

Firstly, we plotted the confusion matrix of all the individual classification models, which is a part of the Model Performance Evaluation,

From Fig. 5a, b, c, d, e, f we calculated performance matrix of each classification models.

In Table 3, Results show that Multinomial Naïve Bayes has the highest accuracy of 84.91% compared to other classification models.

To Improve the accuracy of all individual classification models, we have fine-tuned the parameters of each model to get better accuracy for the comparison.

In Table 4, after hyperparameter tuning of the Decision tree, we can see that the max-depth=9 gave the best accuracy of 73.88%.

In Table 5, after hyperparameter tuning of Multinomial Naïve Bayes, we can see that the alpha=0.9 gave the best accuracy of 84.91%.

In Table 6, after hyperparameter tuning of Random Forest, we can see that the n-estimators=100 gave the best accuracy of 79.08%.

In Table 7 after hyperparameter tuning of SVM, we can see that the Kernel=sigmoid gave the best accuracy of 81.02%.

In Table 8, after hyperparameter tuning of Logistic Regression, we can see that the C=1 gave the best accuracy of 81.02%.

In Table 9, after hyperparameter tuning of Logistic Regression, we can see that the n-neighbors=1 gave the best accuracy of 70.8%.

In Fig. 6, Indicates that before performing Hyperparameter tuning all the six classification models have reached their maximum accuracy, However, Naïve Bayes has got the highest accuracy of 84.91% among the four classification models.

Figure 7, Indicates that after performing Hyperparameter tuning all four classification models with their best parameters have reached their maximum accuracy, Again Naïve Bayes has the highest accuracy of 84.91% with an alpha value of 0.9 among the six classification models.

Further, we have compared different combinations of classification models using a majority voting classifier with a hard voting technique. The combination is based on the models after Hyperparameter Tuning and also the models that gave the best accuracy after Hyperparameter Tuning. The results obtained from the combination selection determine which combination got the highest accuracy.

Firstly, we plotted the confusion matrix of different combinations of individual models where each model has the best parameter obtained from Hyperparameter tuning.

From Fig. 8, we calculated the performance matrix of each Ensemble model.

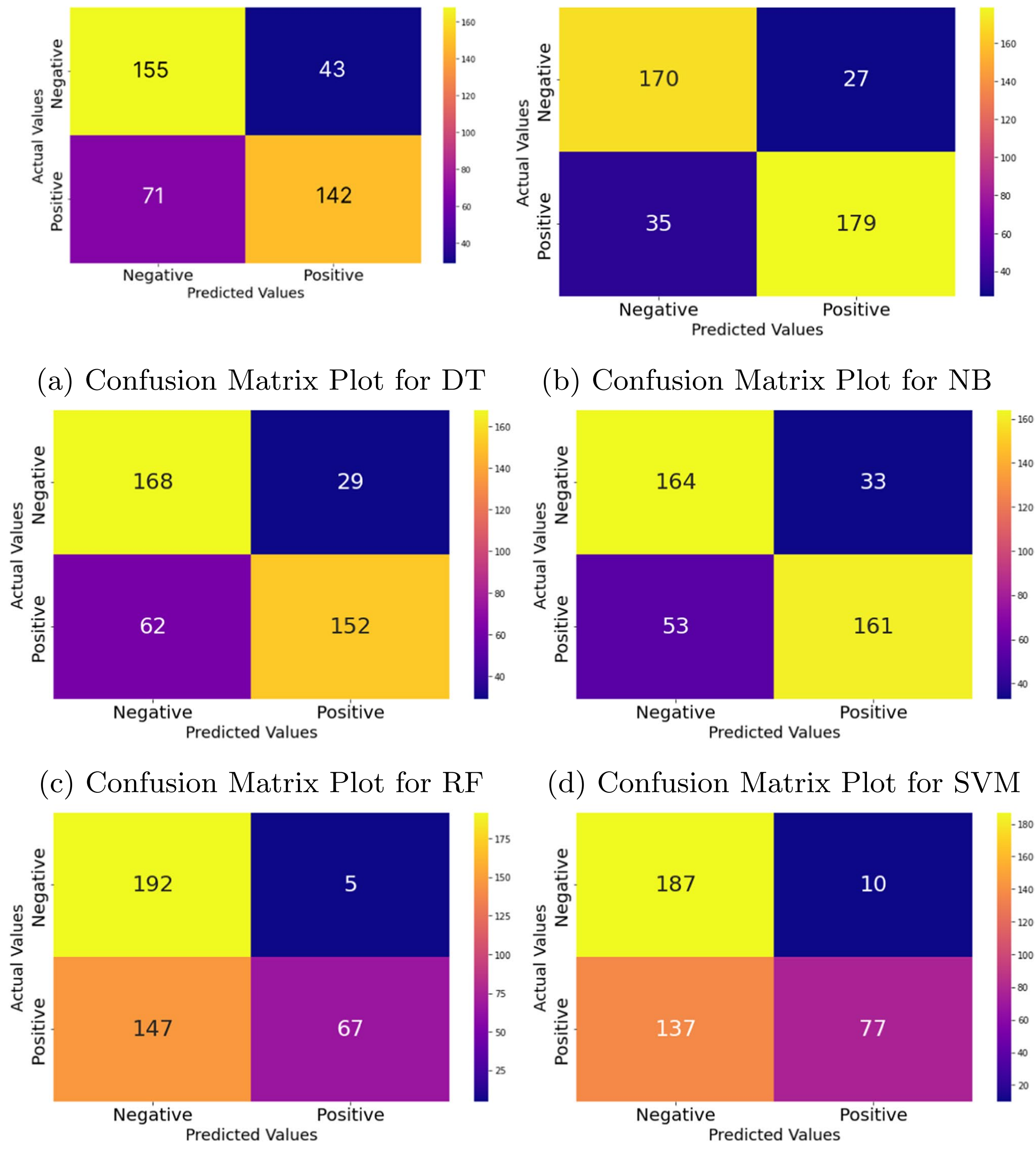
In Table 10, Results show that Combination-2 (NB, SVM, LR) has the highest accuracy of 88.27% compared to other Combination selections.

Furthermore, we have analyzed the processing time of both individual and ensemble models to find the highest accuracy within a short period.

In Table 11, Results show that Combination-3 (NB, SVM) has the best processing time of 3.57 s compared to other models.

From our Experiment referring to Fig. 9, we have found five models in terms of best processing time. Out of five models, one is an individual classification model and the remaining three are ensemble models. In the individual classification model, Multinomial Naïve Bayes has the best processing time of 4.09 s compared to another Individual classification model, In Ensemble Models, the combination of Decision Tree, Multinomial Naïve Bayes, Support Vector Machine, Random Forest, Logistic Regression, and K-Nearest Neighbors (C1) got processing time of 4.17 s, the combination of Multinomial Naïve Bayes, Support Vector Machine and Logistic Regression (C2) got processing time of 4.05 s, the combination of Multinomial Naïve Bayes and Support Vector Machine (C3) got processing time of 3.57 s, the combination of Multinomial Naïve Bayes and Logistic Regression (C3) got processing time of 3.94 s.

Out of these five models we chose the best two models in terms of accuracy and processing time referring to Figs. 9 and 10 i.e., the combination of Multinomial Naïve Bayes, Support Vector Machine and Logistic Regression (C2) got best accuracy of 88.27% with a processing time of 4.05 s and also the combination of Multinomial Naïve Bayes and Support Vector Machine (C3) got best accuracy of 87.37% with a processing time of 3.57 s.



(a) Confusion Matrix Plot for DT (b) Confusion Matrix Plot for NB

(c) Confusion Matrix Plot for RF (d) Confusion Matrix Plot for SVM

(e) Confusion Matrix Plot for LR (f) Confusion Matrix Plot for KNN

Fig. 5 Confusion matrix for all the machine learning models

Even though the C3 i.e., Multinomial Naïve Bayes and Support Vector Machine has the best processing time but lacks slightly in performance (Accuracy) model C2 i.e., Multinomial Naïve Bayes, Support Vector Machine and Logistic Regression has the best performance accuracy and also good processing time.

Table 3 Performance metrics results

S. No.	Model	Accuracy (%)	Precision	Recall	F1-Score
1	Decision Trees	72	0.71	0.66	0.68
2	**Multinomial Naïve Bayes**	**84.91**	**0.87**	**0.84**	**0.85**
3	Random Forest	77.86	0.84	0.71	0.77
4	Support Vector Machine	79.08	0.83	0.75	0.79
5	Logistic Regression	63.02	0.93	0.31	0.47
6	K-Nearest Neighbors	64.23	0.89	0.36	0.51

Bold values indicates better results

Table 4 Hyperparameter tuning results for decision tree

Parameters (max_depth)	Accuracy (%)
1	53.77
2	55.72
3	58.64
4	61.31
5	63.26
6	64.48
7	66.18
8	67.4
9	**73.88**
10	73.64

Bold values indicates better results

Table 5 Hyperparameter tuning results for multinomial naïve bayes

Parameters (alpha)	Accuracy (%)
0.1	83.7
0.2	83.45
0.3	83.94
0.4	83.45
0.5	83.45
0.6	83.94
0.7	84.43
0.8	84.67
0.9	**84.91**
1.0	84.91

Bold values indicates better results

Table 6 Hyperparameter tuning results for random forest

Parameters (n_estimators)	Accuracy (%)
50	78.59
100	**79.08**
150	78.1
200	78.1

Bold values indicates better results

Table 7 Hyperparameter Tuning results for SVM

Parameters (Kernels)	Accuracy (%)
linear	78.08
rbf	80.05
poly	66.67
sigmoid	**81.02**

Bold values indicates better results

Table 8 Hyperparameter tuning results for logistic regression

Parameters (C)	Accuracy (%)
0.001	47.93
0.01	63.02
0.1	77.13
1	**81.02**
10	78.35

Bold values indicates better results

Table 9 Hyperparameter Tuning results for K-Nearest Neighbors

Parameters (n_neighbors)	Accuracy (%)
1	**70.8**
2	59.37
3	64.23
4	57.66
5	62.53
6	56.69
7	61.07
8	55.47
9	59.61
10	55.47
11	56.45
12	55.23
13	58.15
14	54.01
15	57.18
16	53.53
17	56.2
18	53.28
19	54.74
20	52.55

Bold values indicates better results

Here the result obtained from our analysis concludes that C2 i.e., Multinomial Naïve Bayes, Support Vector Machine, and Logistic Regression Ensemble technique is recommended for the analysis of the sentiment of the restaurant reviews than other combinations of Ensemble models and Individual classification Models, offering valuable insights to improve customer satisfaction and business insights for restaurant owners.

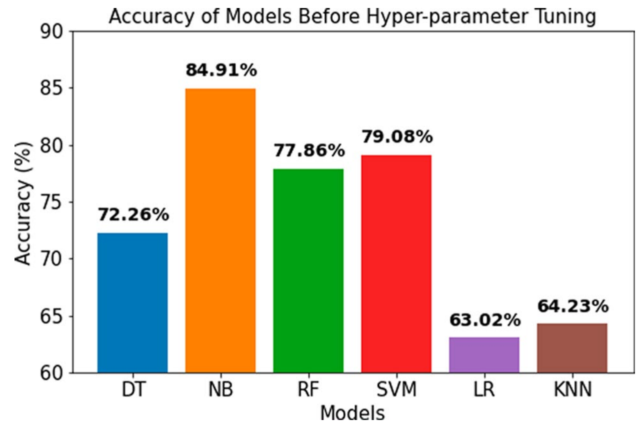


Fig. 6 Accuracy of the models before Hyper-Parameter

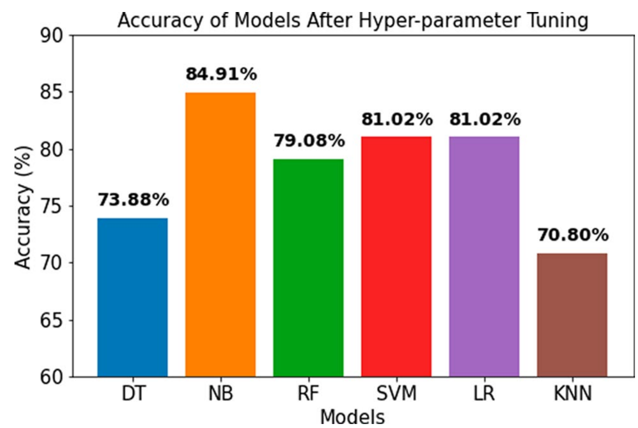


Fig. 7 Accuracy of the models after Hyper-Parameter

5 Comparative analysis

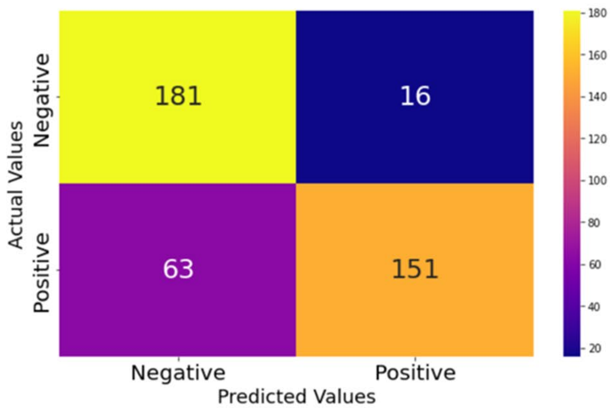
We referred to the article by S. Al-Saqqa et al [25], which applied a set of four models to an Arabic dataset for sentiment analysis. While we drew theoretical insights from their approach, our study diverges technically by applying an ensemble method to a dataset of English restaurant reviews. We utilized a broader range of classification models, both individually and in combination, to achieve higher accuracy. To validate the effectiveness of our proposed method, we conducted comparative analyses with several recent studies.

The authors in article [25] have used three datasets for the opinion corpus. For the Arabic dataset, the combination of Naive Bayes, SVM, and Decision Trees achieved the highest accuracy of 86.8%. For the UCI dataset, SVM achieved the highest accuracy of 84.4%. For the LABR dataset, the combination of Naive Bayes, SVM, and Decision Trees achieved an accuracy of 85.94%. In our study using a restaurant review dataset, the combination of Naive Bayes, SVM, and Logistic Regression achieved the highest accuracy of 88.27% with the best processing time of 3.57 s as shown in Fig. 11.

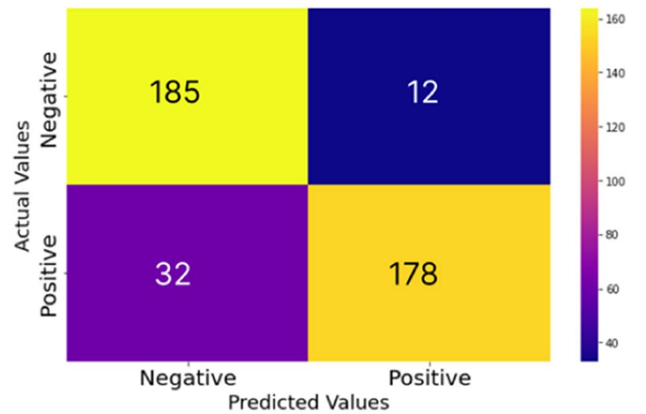
6 Conclusion and future works

This study presents a comprehensive comparison of machine learning models to analyze sentiment in restaurant customer reviews, utilizing a dataset of 1,424 reviews labeled as either positive or negative. The data was thoroughly pre-processed through cleaning, normalization, and tokenization to prepare it for analysis. We evaluated six different classification models and employed hyperparameter tuning to optimize each model's performance.

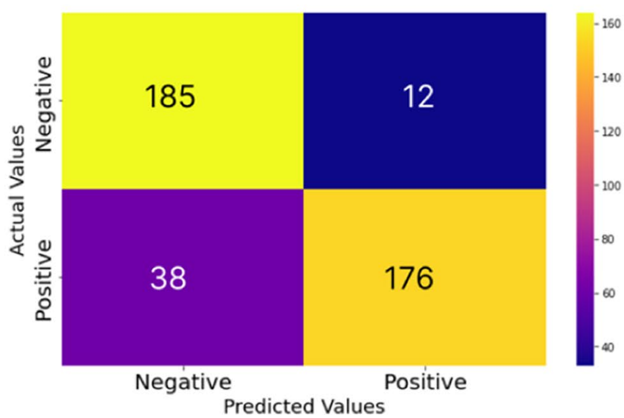
A significant contribution of this study is the application of an ensemble technique using the Majority Voting algorithm, which combined the strengths of various models to enhance classification accuracy and overall performance.



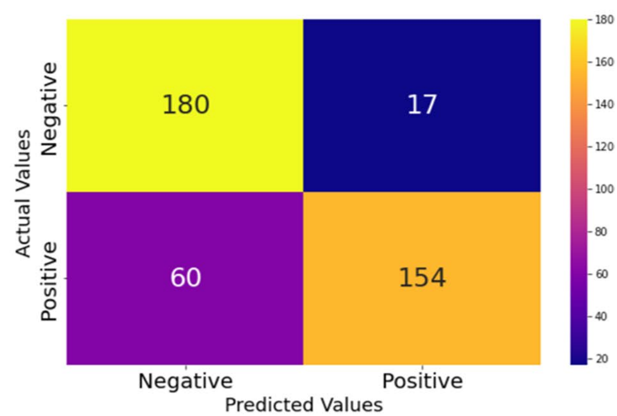
(a) Confusion Matrix Plot for C1



(b) Confusion Matrix Plot for C2



(c) Confusion Matrix Plot for C3



(d) Confusion Matrix Plot for C3

Fig. 8 Confusion matrix for all Combinations (Table 2)

Table 10 Ensemble models performance metrics results

Combinations	Models Combined	Accuracy (%)	Precision	Recall	F1-Score
C1	DT, NB, SVM, RF, LR, KNN	85.75	0.92	0.78	0.84
C2	**NB, SVM, LR**	**88.27**	**0.87**	**0.76**	**0.81**
C3	NB, SVM	87.37	0.9	0.72	0.8
C4	NB, LR	86.51	0.89	0.73	0.81

Bold values indicates better results

We also assessed the processing time of each ensemble model, aiming to identify the best trade-off between accuracy and computational efficiency.

Our findings demonstrate that restaurants can leverage this sentiment analysis approach to gain valuable insights from customer feedback, enabling data-driven decisions that can significantly improve customer satisfaction. By accurately deciphering customer sentiment, businesses can refine their services and offerings, ultimately leading to increased customer loyalty and business success.

To further improve the effectiveness of sentiment analysis in this context, future research should focus on expanding the dataset to include a more diverse range of reviews from various demographics and restaurant types. Additionally, exploring advanced machine learning techniques such as Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), and BERT could provide deeper insights and enhance the accuracy of sentiment analysis. These approaches hold the potential to push the boundaries of current methodologies, offering even more precise and

Table 11 Processing time results

S. No.	Models	Processing Time (in Sec)
1	Decision Trees	5.44
2	Multinomial Naïve Bayes	4.09
3	Random Forest	10.07
4	Support Vector Machine	7.02
5	Logistic Regression	6.26
6	K-Nearest Neighbors	7.35
7	C1(DT, NB, SVM, RF, LR, KNN)	4.17
8	C2(NB, SVM, LR)	4.05
9	**C3(NB, SVM)**	**3.57**
10	C4(NB, LR)	3.94

Bold values indicates better results

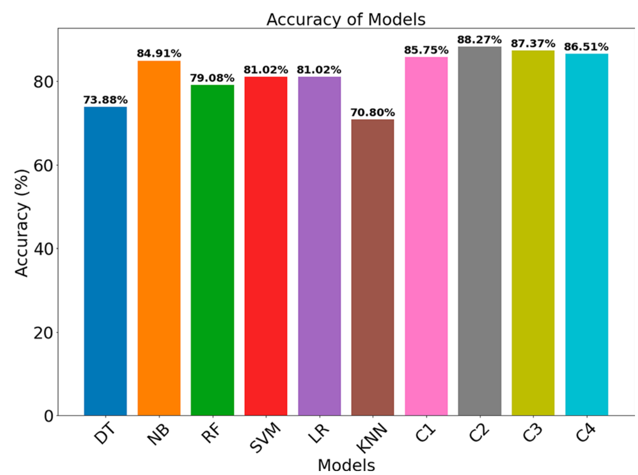


Fig. 9 Results of Individual Models for Accuracy

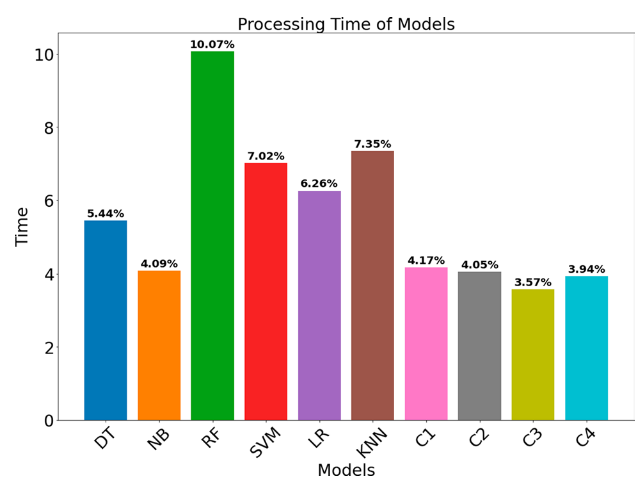


Fig. 10 Time taken by the individual Models (Processing Time)

actionable results for the restaurant industry. The model struggles to accurately interpret and classify sarcastic comments, affecting overall sentiment prediction accuracy.

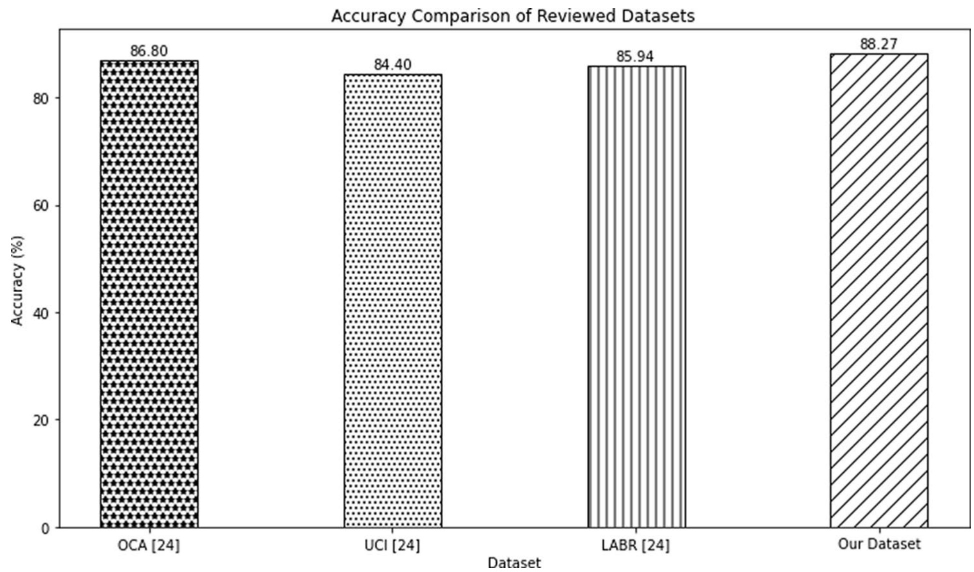


Fig. 11 Comparing our model with the existing works [25]

Acknowledgements Not Applicable.

Author contributions All authors contributed to the study conception and design. Rohan Kamath and Kaushik M R implemented, wrote, and reviewed the manuscript and Ramakrishna M reviewed and supervised the manuscript work. The first draft of the manuscript was written by Rohan Kamath and Kaushik M R and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding Not Applicable.

Data availibility The data set considered in the research works is publicly available.

Declarations

Ethics approval anf condent to participate Not Applicable.

Research involving human participants and/or animals Not Applicable

Informed consent Not Applicable.

Competing interests On behalf of all authors, the corresponding author states that there is no competig interets

References

1. Wankhade M, Rao ACS, Kulkarni C. A survey on sentiment analysis methods, applications, and challenges. Artif Intell Rev. 2022;55(7):5731–80.
2. Medhat W, Hassan A, Korashy H. Sentiment analysis algorithms and applications: a survey. Ain Shams Eng J. 2014;5(4):1093–113.
3. Tiwari D, Nagpal B, Bhati BS, Mishra A, Kumar M. A systematic review of social network sentiment analysis with comparative study of ensemble-based techniques. Artif Intell Rev. 2023;56:13407–61.
4. Nielsen FÅ. "A new ANEW: evaluation of a word list for sentiment analysis in microblogs. CEUR Workshop Proc. 2011;93:98.
5. Elbagir S, Yang J. "Twitter sentiment analysis using natural language toolkit and vader sentiment," in Proceedings of the international multiconference of engineers and computer scientists, 2019;122:16.

6. Jonathan B, Sihotang JI, Martin S. Sentiment analysis of customer reviews in zomato bangalore restaurants using random forest classifier. Abstract Proc Int Scholars Conf. 2019;7:1719–28.
7. Chua LO, Roska T. The CNN paradigm. IEEE Trans Circuits Syst I: Fund Theory Appl. 1993;40(3):147–56.
8. Li S, Li W, Cook C, Zhu C, Gao Y. "Independently recurrent neural network (indrnn): Building a longer and deeper rnn," in Proceedings of the IEEE conference on computer vision and pattern recognition, 2018;5457–5466.
9. Jelodar H, Wang Y, Yuan C, Feng X, Jiang X, Li Y, Zhao L. Latent Dirichlet allocation (LDA) and topic modeling: models, applications, a survey. Multimed Tools Appl. 2019;78:15169–211.
10. Hoang M, Bihorac OA, Rouces J. "Aspect-based sentiment analysis using bert," in Proceedings of the 22nd nordic conference on computational linguistics, 2019;187–196.
11. Liu X, Zheng Y, Du Z, Ding M, Qian Y, Yang Z, Tang J. "Gpt understands, too," AI Open, (2023).
12. Patil A, Bheda K, Upadhyay NS, Sawant R. "Restaurant's feedback analysis system using sentimental analysis and data mining techniques," in 2018 International Conference on Current Trends towards Converging Technologies (ICCTCT), 2018;1–4, IEEE.
13. Bhavitha B, Rodrigues AP, Chiplunkar NN. "Comparative study of machine learning techniques in sentimental analysis," in 2017 International conference on inventive communication and computational technologies (ICICCT), 2017;216–221, IEEE.
14. Rahate S, Dehanka V, Teppalwar T, Surjuse V. Review sentimental analysis. Int J Comput Sci Mobile Comput. 2022;11:37–41.
15. Krishna A, Akhilesh V, Aich A, Hegde C. "Sentiment analysis of restaurant reviews using machine learning techniques. Cham: Springer; 2018.
16. Laksono RA, Sungkono KR, Sarno R, Wahyuni CS. "Sentiment analysis of restaurant customer reviews on tripadvisor using naïve bayes," in 2019 12th international conference on information & communication technology and system (ICTS), 2019;49–54, IEEE.
17. Shamantha RB, Shetty SM, Rai P. "Sentiment analysis using machine learning classifiers: evaluation of performance," in 2019 IEEE 4th international conference on computer and communication systems (ICCCS), 2019;21–25, IEEE.
18. Zahoor K, Bawany NZ, Hamid S. "Sentiment analysis and classification of restaurant reviews using machine learning," in 2020 21st International Arab Conference on Information Technology (ACIT), 2020;1–6, IEEE.
19. Sharif O, Hoque MM, Hossain E. "Sentiment analysis of bengali texts on online restaurant reviews using multinomial naïve bayes," in 2019 1st international conference on advances in science, engineering and robotics technology (ICASERT), 2019;1–6, IEEE.
20. Huda SA, Shoikot MM, Hossain MA, Ila IJ. "An effective machine learning approach for sentiment analysis on popular restaurant reviews in bangladesh," in 2019 1st International Conference on Artificial Intelligence and Data Sciences (AiDAS), 2019;170–173, IEEE.
21. Gunawan L, Anggreainy MS, Wihan L, Lesmana GY, Yusuf S, et al. Support vector machine based emotional analysis of restaurant reviews. Proc Comput Sci. 2023;216:479–84.
22. Alrehili A, Albalawi K. "Sentiment analysis of customer reviews using ensemble method," in 2019 International conference on computer and information sciences (ICCIS), 2019;1–6, IEEE.
23. Mamun MMR, Sharif O, Hoque MM. Classification of textual sentiment using ensemble technique. SN Comput Sci. 2022;3(1):49.
24. Oussous A, Lahcen AA, Belfkih S. "Improving sentiment analysis of moroccan tweets using ensemble learning. Cham: Springer; 2018.
25. Al-Saqqa S, Obeid N, Awajan A. "Sentiment analysis for arabic text using ensemble learning," in 2018 IEEE/ACS 15th International Conference on Computer Systems and Applications (AICCSA), 2018;1–7, IEEE.
26. Carrasco P, Dias S. Enhancing restaurant management through aspect-based sentiment analysis and NLP techniques. Proc Comput Sci. 2024;237:129–37.
27. Gupta V, Rattan P. Advancing sentiment analysis in restaurant reviews through unsupervised machine learning algorithms. Int J Intell Eng Syst. 2024;17:4.
28. "https://www.kaggle.com/,"
29. Somvanshi M, Chavan P, Tambade S, Shinde SV. "A review of machine learning techniques using decision tree and support vector machine," in 2016 International Conference on Computing Communication Control and automation (ICCUBEA), 2016;1–7.
30. Su J, Shirab JS, Matwin S. "Large scale text classification using semi-supervised multinomial naive bayes," in Proceedings of the 28th international conference on machine learning (ICML-11), 2011;97–104.
31. Rakotomamonjy A. Variable selection using SVM-based criteria. J Machine Learn Res. 2003;3:1357–70.
32. Evans JS, Murphy MA, Holden ZA, Cushman SA. Modeling species distribution and change using random forest. Cham: Springer; 2010.
33. Peng C-YJ, So T-SH, Stage FK, St. John EP. 2002"The use and interpretation of logistic regression in higher education journals 1988–1999. Research in higher education. 43:259–293.
34. Chirici G, Mura M, McInerney D, Py N, Tomppo EO, Waser LT, Travaglini D, McRoberts RE. A meta-analysis and review of the literature on the k-nearest neighbors technique for forestry applications that use remotely sensed data. Remote Sens Environ. 2016;176:282–94.
35. Ruta D, Gabrys B. Classifier selection for majority voting. Information fusion. 2005;6(1):63–81.